

Extra Practice Problems on L^{-1}
Transform Calculus

Koeli Ghoshal

1. Find $L^{-1} \left\{ \frac{5s+4}{s^3} - \frac{2s-18}{s^2+9} + \frac{24-30\sqrt{s}}{s^4} \right\}$

Ans: $5t + 2t^2 - 2\cos 3t + 6\sin 3t + 4t^3 - 16t^{5/2}/\sqrt{\pi}$

2. Find $L^{-1} \left\{ \frac{1}{\sqrt{2s+3}} \right\}$ Ans: $\frac{1}{\sqrt{2\pi}} t^{-1/2} e^{-3t/2}$

3. Find $L^{-1} \left\{ \frac{(s+1)e^{-\pi s}}{s^2+s+1} \right\}$

Ans: $\frac{e^{-\frac{1}{2}(t-\pi)}}{\sqrt{3}} \left\{ \sqrt{3}\cos\frac{\sqrt{3}}{2}(t-\pi) + \sin\frac{\sqrt{3}}{2}(t-\pi) \right\} u(t-\pi)$

4. Find $L^{-1} \left\{ \frac{se^{-\frac{4\pi s}{5}}}{s^2+25} \right\}$ Ans: $\cos 5t u(t-4\pi/5)$

5. Find $L^{-1} \left\{ \frac{1}{s^2(s+1)^2} \right\}$ Ans: $te^{-t} + 2e^{-t} + t - 2$

6. Find $L^{-1} \left\{ \frac{e^{4-3s}}{(s+4)^{5/2}} \right\}$ Ans: $\frac{4(t-3)^{3/2} e^{-4(t-4)}}{3\sqrt{\pi}} u(t-3)$

7. Find $L^{-1} \left\{ \frac{s}{(s^2+a^2)^2} \right\}$ Ans: $\frac{t \sin at}{2a}$

8. Show that $L^{-1} \left\{ \frac{1}{s} \ln \left(\frac{s+2}{s+1} \right) \right\} = \int_0^t \frac{e^{-u} - e^{-2u}}{u} du$

9. Show that $L^{-1} \left\{ \frac{1}{s} \ln \left(\frac{s^2+a^2}{s^2+b^2} \right) \right\} = \int_0^t \frac{\cos au - \cos bu}{u} du$

10. Find $L^{-1} \left\{ \frac{s}{(s^2+4)^3} \right\}$ Ans: $\frac{1}{64} t (\sin 2t - 2t \cos 2t)$

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